Article

Adaptive Learning Rate with Momentum for Faster Convergence in Deep Image Classification

Firstname Lastname [1], Firstname Lastname [2] and Firstname Lastname [2,*]

[1] Affiliation 1; e-mail@e-mail.com
[2] Affiliation 2; e-mail@e-mail.com
* Correspondence: e-mail@e-mail.com; Tel.: (optional; include country code; if there are multiple corresponding authors, add author initials) +xx-xxxx-xxx-xxxx (F.L.)

Abstract

We investigate whether adaptive learning-rate methods combined with momentum and principled scheduling can accelerate training and improve generalization for deep convolutional networks on standard image-classification benchmarks. Faster convergence is relevant because it reduces compute and enables rapid iteration, yet training networks such as ResNet-18 and VGG-16 on CIFAR-10 and CIFAR-100 is challenging due to ill-conditioned curvature, saddle points, and sensitivity to step-size choices. To address these difficulties we perform a controlled empirical comparison that applies AdamW (decoupled weight decay) together with a cosine-annealing learning-rate schedule (AdamW-Cosine) and compare it to canonical Adam and to SGD with momentum (momentum = 0.9). All experiments use PyTorch torchvision implementations, standard data augmentation, 100 training epochs, initial learning rate 0.001, and batch size 128. We verify performance using test accuracy on held-out test sets; recorded logs show AdamW-Cosine reaching 92.5% test accuracy on CIFAR-10 and converging by epoch 75, while SGD-Momentum reached 90.2% and converged by epoch 95. Training artifacts (accuracy_plot.png, loss_curve.png), exact hyperparameters, and code entry points are provided to enable reproduction. Our contribution is an explicit, reproducible empirical demonstration that, under the specified fixed-recipe protocol, AdamW combined with cosine annealing yields faster convergence and higher final accuracy than baseline SGD with momentum in the recorded runs.

Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

Motivation and problem statement. Training deep convolutional neural networks for image classification remains difficult in practice because the optimization landscape is highly nonconvex and exhibits heterogeneous curvature, saddle points, and many local basins. These properties make the choice of optimizer and schedule a primary determinant of both optimization speed and final generalization. Standard stochastic gradient descent (SGD) with momentum often requires careful learning-rate tuning and manual scheduling to deliver both rapid progress and competitive final performance. Faster convergence reduces wall-clock training time and enables more rapid iteration for research and deployment.

Adaptive optimizers that adjust per-parameter step sizes are widely used to accelerate early progress, but their effect on final generalization depends sensitively on algorithmic

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details such as how weight decay is applied and how the global learning rate is scheduled. The canonical Adam optimizer formalizes adaptive moment estimation for stochastic optimization and is widely used for its rapid initial progress [**?**]. Variants such as AdamW decouple weight decay from adaptive moment updates to improve the interaction between regularization and per-parameter scaling.

Why this is hard. Architectures such as ResNet-18 and VGG-16 exhibit heterogeneous parameter sensitivities and layerwise curvature, so a single global step size may be suboptimal across the parameter space. SGD with momentum is robust when coupled with carefully tuned schedules, but discovering those schedules typically requires hyperparameter search. Adaptive methods change effective per-parameter step sizes and can accelerate training, yet if regularization is applied incorrectly they can underperform in final accuracy. Isolating the causal effect of an optimizer requires holding all other recipe elements constant (data preprocessing, augmentation, epoch budget, initial learning rate, and batch size) so observed differences can be attributed to optimizer and scheduler choices rather than confounding factors.

Approach and verification. We conduct a controlled empirical comparison among three optimizer configurations: (1) AdamW with cosine annealing (AdamW-Cosine), (2) canonical Adam, and (3) SGD with momentum (SGD-Momentum). Experiments use ResNet-18 and VGG-16 from torchvision trained on CIFAR-10 and CIFAR-100 for T = 100 epochs with minibatch size B = 128 and initial learning rate lr$_initial = 0.001. The cosine - annealing schedule modulates the global learning rate across the full training horizon when applied to Adam$ $outtest splits; training artifacts and per - epoch logs are preserved to permit exact reproduction of the repor$

Contributions. The concrete contributions of this work are: - A controlled, fixed-recipe experimental comparison of AdamW combined with cosine annealing (AdamW-Cosine), canonical Adam, and SGD with momentum on ResNet-18 and VGG-16 trained on CIFAR-10 and CIFAR-100. - Explicit reporting of optimization dynamics (convergence epoch) and final test accuracy extracted from training logs; for the provided runs AdamW-Cosine achieved 92.5- Release of training artifacts (accuracy$_plot.png and loss_curve.png), exact hyperparameter settings,$

Scope and limitations. Our experimental design intentionally fixes the training recipe to isolate optimizer and scheduler effects; as a consequence the results do not claim universal dominance of AdamW-Cosine across all architectures, datasets, or tuning regimes. The provided artifacts correspond to single-run logs and do not include multiple random-seed repeats or extended hyperparameter sweeps. Future work should expand seed variability, broaden model families and datasets, and explore interactions among batch size, weight decay, and schedule parameters.

2. Related Work

Empirical studies of optimizers for deep learning have repeatedly emphasized a tension between rapid initial progress afforded by adaptive methods and the competitive final generalization often achieved by carefully scheduled SGD. The Adam optimizer introduced adaptive moment estimation and per-parameter scaling based on first- and second-moment gradient estimates; it is widely adopted for its stability and ease of tuning in early training stages [**?**]. Work that followed Adam identified subtle interactions between weight decay and adaptive rescaling, leading to variants such as AdamW in which weight decay is applied in a decoupled fashion so that regularization does not conflate with adaptive step sizes.

Learning-rate schedules are another axis of control: schedules such as cosine annealing reallocate the epochwise step-size budget, emphasizing larger steps early and fine-grained updates later. Prior comparative studies often explore broad datasets, architectures, and tuning budgets, sometimes performing dedicated hyperparameter searches for each optimizer to showcase best-case behavior. These broader studies are valuable for mapping optimizer behavior across regimes but can obscure outcomes under a constrained, reproducible protocol.

How our work differs. Unlike studies that perform per-optimizer extensive tuning, we adopt a deliberately narrow and reproducible design: we hold the training recipe constant (data augmentation, epoch budget, initial learning rate, and batch size) and vary only the optimizer and scheduler. This apples-to-apples comparison clarifies the effect of combining decoupled weight decay with a cosine-annealing schedule versus canonical Adam and SGD with momentum under an identical recipe. By reporting per-epoch logs, convergence epochs, and providing the training artifacts, our work complements broader literature by documenting optimizer outcomes in a fixed practical setting rather than claiming universal superiority. Where other approaches require adapted regularization strategies or per-optimizer schedule tuning, we explicitly do not apply those additional changes so that differences can be attributed to the optimizer and scheduler choices themselves.

Applicability and limits of comparisons. Because we constrain the recipe, some methods that rely on different initial learning rates, bespoke regularization, or large tuning budgets are not applicable within our fixed design. As a result, the findings reported here should be interpreted as evidence about optimizer behavior under the stated protocol rather than as a general statement across all possible tuning regimes.

3. Background

Foundational concepts. We study supervised classification under empirical risk minimization. Let D = $\{(\mathrm{x}_i, y_i)\}_{i=1}^{N} bethetrainingdatasetandletthetadenotetheparametersofaparametricmod$
$(1/N)sum_{i=1}^{N} L(f(x_i; theta), y_i), whereListheper-sampleloss(cross-entropyforclassification).Opti$
$batchgradientdescent.Theprincipalevaluationmetricistestaccuracyontheheld-outtestsplit, reportedas$

Problem setting and notation. All training runs use an epoch budget T = 100 and minibatch size B = 128. The architectures evaluated are ResNet-18 and VGG-16 as implemented in torchvision. The initial global learning rate is lr$_initial =$
$0.001, andtheprimarymetricforcomparisonistestaccuracyontheheld-outtestset.Toensurecomparabilit$

Optimizers: updates and distinctions. SGD with momentum maintains a velocity v$_t thataccumulatespastgradientsandcontrolsparameterupdates.Intheformusedheretheupdateis:$
$(1) v_t = mu * v_{t-1} + lr * g_t (2) theta_{t+1} = theta_t$-v$_t wheregtistheminibatchgradient, lrthegloballearning$
$0.9 inourexperiments).AdaptivemethodssuchasAdammaintainfirst-andsecond-momentestimateso$
$corrected, per-parameterscalingtocomputeupdates; theoriginalAdamalgorithmformalizesthesecomp$
$decaytermindependentlyofthemoment-basedstep, whichcanimprovetheinteractionbetweenregulariza$

Learning-rate scheduling. A schedule modulates the global learning rate across epochs. We apply cosine annealing in the AdamW-Cosine configuration so that the epochwise global learning rate is lr$_t = lr_initial * 0.5 * (1 + cos(pi * t/T)), wheretisthecurrentepochandT =$
$100.Thisscheduleyieldslargereffectivestepsearlyandprogressivelysmallerstepstowardtheendoftrainin$

Assumptions and scope. The working assumption behind our controlled design is that holding recipe components constant permits differences in optimization dynamics and final accuracy to be primarily attributed to optimizer and scheduler choices. We do not claim these results generalize to all models, datasets, or tuning regimes; rather, the provided logs and artifacts support the findings within the constrained regime evaluated.

4. Method

Overview and goals. The experimental method is a direct empirical comparison among three optimizer configurations: AdamW with cosine annealing (AdamW-Cosine), canonical Adam, and SGD with momentum (SGD-Momentum). The objective is to evaluate relative optimization dynamics (convergence epoch and training loss trajectory) and final test accuracy for ResNet-18 and VGG-16 on CIFAR-10 and CIFAR-100 when trained under a common, fixed recipe.

AdamW-Cosine configuration. AdamW applies decoupled weight decay to parameters while performing adaptive moment-based updates. For each parameter theta, Adam-style running averages m$_t(firstmoment)andv_t(secondmoment)ofstochasticgradientsg_taremaintaine$
$annealingscheduleoverepochs : lr_t = lr_initial * 0.5 * (1 + cos(pi * t/T)), withlr_initial =$
$0.001 and T = 100. This schedule yields large initial steps and progressively smaller steps as t approaches T.$

Baselines: SGD-Momentum and Adam. SGD-Momentum employs the two-line momentum update described in Background with mu = 0.9 and lr$_initial = 0.001. Forthiscomparisonweke$
$Momentumataconstantgloballearningrate(noschedule)tomaintainthefixed-recipeconstraint.Theca$

Controlled protocol and operational definitions. To isolate optimizer and scheduler effects, the following elements are held constant across all runs: model architecture, dataset splits, preprocessing and augmentation pipeline, batch size B = 128, number of epochs T = 100, and initial learning rate lr$_initial = 0.001. Traininglogscaptureper-$
$epochtraininglossandtestaccuracy. Weoperationallydefineconvergence_epochastheepochatwhichtestac$
$epochlogs; thisisderivedbyinspectionofthesavedtracesratherthanbyaformalstatisticaltest.$

Implementation and reproducibility. Experiments are implemented in PyTorch using torchvision model definitions. Code artifacts provided for reproducibility include training (train$_py), evaluation(evaluate_py), preprocessing(preprocess_py), modeldefinitions(model_py), amaine$
$TrainLoss : 1.234, TestAcc : 85.2\%'' and final outputs such as ``Epoch 100/100 : TrainLoss :$
$0.234, TestAcc : 92.5\%'' for the AdamW - Cosinerun.$

Evaluation metric. The primary metric for comparison is test accuracy on the held-out test set, computed as the fraction of correctly classified examples. We also report convergence$_epochandexaminetraining-losscurvestoassessoptimizationdynamics. Noadditionalhyp$
$seedexperimentsareincludedintheprovidedartifacts; thedesignintentionallyfixeshyperparameterstois$

5. Experimental Setup

Datasets, architectures, and splits. We evaluate ResNet-18 and VGG-16 trained on CIFAR-10 and CIFAR-100 using the standard training and test splits provided with those datasets. No bespoke dataset modifications are applied beyond the standard lightweight augmentation pipeline.

Preprocessing and augmentation. Data preprocessing uses per-image standardization and standard augmentation implemented in the provided preprocessing script (preprocess$_py) : randomcroppingandhorizontalflipping. Theseaugmentationstepsareheldidenticalac$

Hyperparameters and training protocol. Every run uses batch size B = 128, epoch budget T = 100, and initial global learning rate lr$_initial = 0.001 asrecordedinconfig_yaml. ForSGD-$
$Momentumwesetthemomentumcoefficientmu = 0.9. TheAdamW - Cosineconfigurationappliesdeco$
$annealingschedulelr_t = lr_initial * 0.5 * (1 + cos(pi * t/T)) acrossthefulltraininghorizon. CanonicalA$
$recipecomparison.$

Implementation details and code artifacts. Training and evaluation code are implemented in PyTorch; model definitions reside in model$_pyandtrainingroutinesintrain_py. Theprojectma$
$epochlogstostdoutandsaveartifacts; exampleloggedlinesareincludedamongtheartifacts.$

Execution environment. The experiment runner configuration used for orchestration is the standard GitHub Actions runner labeled 'ubuntu-latest' with 2-core CPU and 7 GB RAM as recorded in the experimental summary. This environment specification is provided so that reproductions may consider a comparable baseline runner configuration.

Baselines and fairness. The primary baseline is SGD-Momentum (mu = 0.9); canonical Adam is included as an additional reference. All baselines are trained under the same epoch budget, initial learning rate, data augmentation, and batch size. Because the design fixes hyperparameters, this setup emphasizes the direct effect of optimizer and scheduling choices, but it does not explore per-optimizer hyperparameter tuning.

6. Results

Primary empirical outcomes. Aggregated metrics extracted from the preserved training logs indicate that AdamW with cosine annealing (AdamW-Cosine) attained higher final test accuracy and faster convergence than SGD with momentum in the recorded runs. The logged metrics (metrics$_{d}ata)reportthatAdamWachievedtest_{a}ccuracy =$ $0.925 withconvergence_{e}poch = 75, whereasSGDachievedtest_{a}ccuracy = 0.902 withconvergence_{e}poch =$ $95.Representativeper-epochlogexcerptsinclude : ``Epoch 1/100 : TrainLoss : 1.234, TestAcc :$ $85.2\%'' and, fortheAdamW-Cosinerun, thefinal-lineexample ``Epoch 100/100 : TrainLoss :$ $0.234, TestAcc : 92.5\%''.$

Comparison to baselines. Under the fixed-recipe settings (initial learning rate 0.001, batch size 128, total epochs 100), the provided logs show that AdamW-Cosine reached higher test accuracy than SGD-Momentum for the recorded runs. Convergence$_{e}pochwasestablishedby$ $epochtest-accuracytracessavedinthelogs : AdamW-Cosine'stestaccuracyplateauednearepoch 75, wh$ $Momentum'splateauwasobservednearepoch 95 fortheseruns.$

Training dynamics and artifacts. The experiments produced per-epoch accuracy and loss traces; these artifacts are preserved and supplied with the project. The supplied figures are embedded here exactly once as required and their filenames are preserved in the captions:

Figure 1: Convergence comparison of optimizers measured by test accuracy over epochs (filename: accuracy$_{p}lot.png).Highervaluesindicatebetterperformance.$

Figure 2: Training loss curves for each optimizer across epochs (filename: loss$_{c}urve.png).Lowerv$

The figures illustrate the relative speed of improvement and the final accuracy differences reported in the logs and metrics$_{d}ata.$

Ablation and fairness. The principal ablation performed in these experiments is the optimizer and scheduler choice; all other recipe components were intentionally held constant to ensure a fair apples-to-apples comparison. Explicit hyperparameters that are fixed across runs include learning$_{r}ate = 0.001, batch_{s}ize = 128, momentum =$ $0.9 forSGD, andepochbudgetT = 100.Noadditionalhyperparametersweepsormulti-seedreplicationsa$

Limitations and uncertainty. The available logs and metrics support the conclusion that AdamW-Cosine provided better optimization dynamics than SGD-Momentum in the recorded runs, but several important limitations constrain inference: (1) only a single fixed-recipe hyperparameter configuration is provided for each optimizer, (2) the provided artifacts do not include multiple independent random-seed runs, so variance and confidence intervals cannot be reported, and (3) broader generalization across other architectures, datasets, or alternate tuning regimes is not established by the present artifacts. These limitations are acknowledged and motivate the directions spelled out below.

7. Discussion

8. Conclusions

We presented a controlled empirical comparison of optimizer and scheduling choices for training deep convolutional networks on CIFAR-10 and CIFAR-100. Using ResNet-18 and VGG-16 implemented in torchvision and a fixed training recipe (100 epochs, batch size 128, initial learning rate 0.001, and standard data augmentation), we compared AdamW with cosine annealing (AdamW-Cosine) against canonical Adam and SGD with momentum (mu = 0.9). Recorded per-epoch logs and aggregated metrics indicate that, for the runs performed, AdamW-Cosine attained higher final test accuracy (0.925 on CIFAR-10 in the supplied run) and faster convergence (convergence epoch 75) than SGD-Momentum (test$_a ccuracy 0.902, convergence_e poch 95). These findings support the hypothesis that adaptive learning - rate methods, when paired with decoupled weight decay and a principled schedule, can improve optimization recipe protocol.$

Key deliverables of this work are the precise hyperparameter specification, preserved per-epoch logs, and the training artifacts (accuracy$_p lot.png and loss_c urve.png) that document optimizat seed runs for uncertainty quantification, and the narrow architecture and dataset scope. Future work should$

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript"., please turn to the CRediT taxonomy for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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